

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie – AI-Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Gemini calls are synchronous inside async API endpoints	A slow request can block others, so response times grow with concurrent users	High
2	Local Flan-T5 model and PyTorch are loaded at startup	Slow start, high memory use and a heavy deployment footprint	Medium
3	No user accounts, database, learning history or automated tests	No personalization or progress tracking; regressions are harder to catch	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Unicorn process running locally	Deploy to a cloud host, run multiple workers behind a load balancer and use the async Gemini client
2	Data Storage	Stateless; nothing is stored	Add a database (SQLite/PostgreSQL) for users, learning history, quiz scores and saved learning paths
3	Performance	Every request calls Gemini; local model loads at import	Cache repeated questions, lazy-load or drop the local model, and stream long responses
4	Security	API key in .env; CORS limited to localhost origins	Add authentication, rate limiting, input length limits, HTTPS and a secrets manager

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User login, learning history and quiz score tracking with a database	1–2 months	Personalized learning and visible progress

Phase 3	Cloud deployment with caching, async calls and rate limiting; automated tests	2–3 months	Supports many concurrent students reliably
Phase 4	Multi-language support, voice input, adaptive quizzes and a teacher dashboard	4–6 months	Wider reach and a fuller learning platform